

Data Mining HW2 Report

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Problem 1. How do you select features for your model input, what preprocessing did you do ?

- 因為 Machine Learning 的特性，能夠選用的 features 越多通常效果會越好，所以我使用所有能夠取得的 features 作為 model input
- 根據 Attributed Description, 我們可以將 attributes 分為以下兩種
 - Scalar - 單一連續的數值，對於此類 attribute 我們不需要額外去做處理
 - age
 - fnlwgt
 - education-num
 - capital-gain
 - capital-loss
 - hours-per-week
 - Category - 以不同字串表示不同種類，對於此類 attribute 需要將它轉成 one hot encoding，以便後續 model 能夠 inferencing
 - workclass
 - education
 - marital-status
 - occupation
 - relationship
 - race
 - gender

- native-country

- Before one hot encoding

	age	workclass	fnlwgt	education	educational-num	marital-status	occupation	relationship	race	gender	capital-gain	capital-loss	hours-per-week	native-country	income
0	52	Private	210736	Some-college	10	Married-civ-spouse	Exec-managerial	Husband	White	Male	3103	0	55	United-States	>50K
1	62	Private	209844	Some-college	10	Divorced	Adm-clerical	Unmarried	White	Female	0	0	30	United-States	<=50K
2	25	Private	410240	HS-grad	9	Never-married	Craft-repair	Own-child	White	Male	0	0	40	United-States	<=50K
3	28	Private	90547	HS-grad	9	Married-civ-spouse	Other-service	Wife	Black	Female	0	0	23	United-States	<=50K
4	28	Private	132326	HS-grad	9	Never-married	Adm-clerical	Own-child	White	Male	0	0	40	United-States	<=50K

- After one hot encoding

	age	fnlwgt	educational-num	capital-gain	capital-loss	hours-per-week	workclass_?	workclass_Federal-gov	workclass_Local-gov	workclass_Never-worked	...	country_Puerto-Rico
0	52	210736	10	3103	0	55	0	0	0	0	...	0
1	62	209844	10	0	0	30	0	0	0	0	...	0
2	25	410240	9	0	0	40	0	0	0	0	...	0
3	28	90547	9	0	0	23	0	0	0	0	...	0
4	28	132326	9	0	0	40	0	0	0	0	...	0

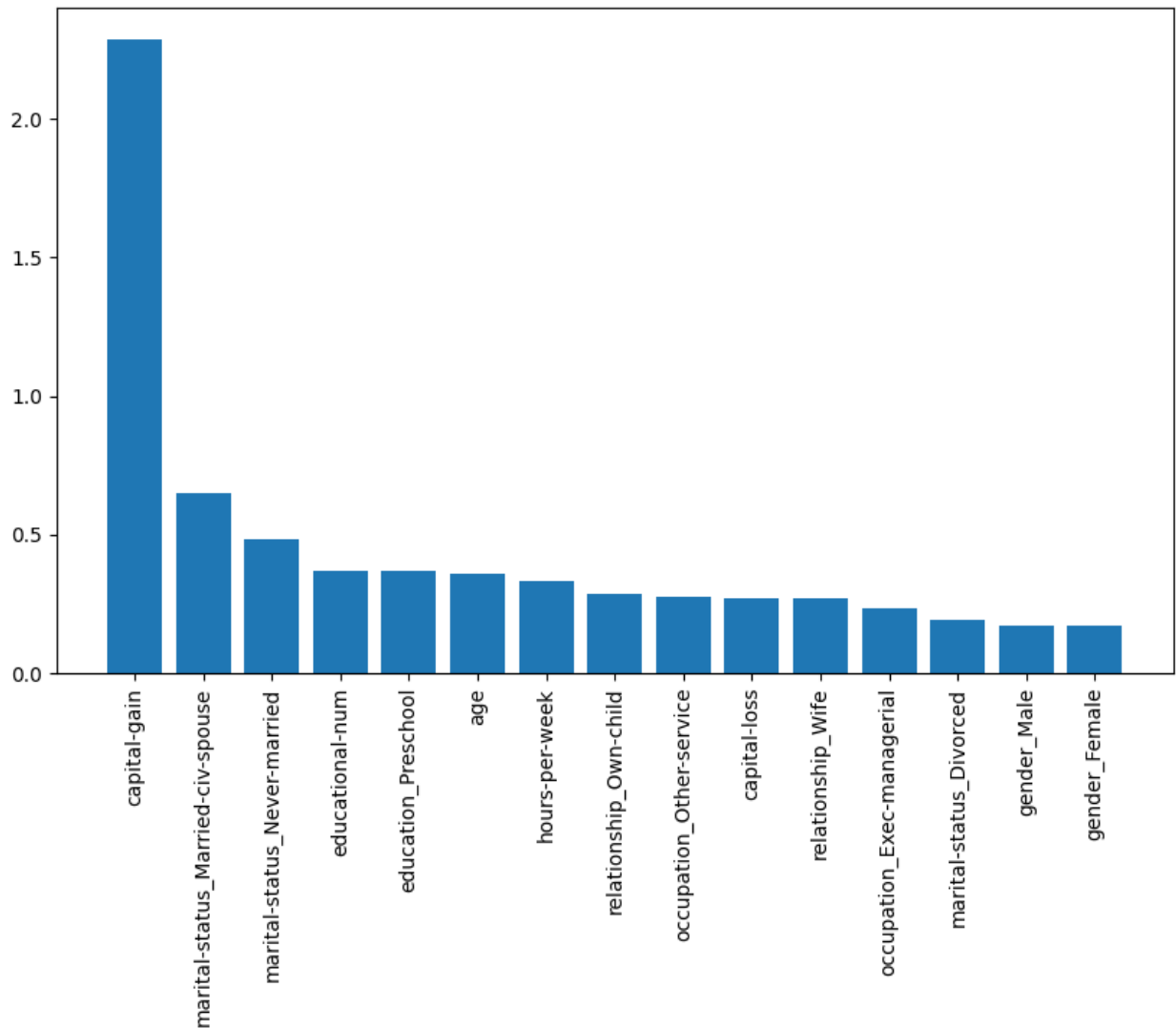
- 為了使模型訓練更加順利，我還額外做了 standarization

```
mean = trainX.mean(axis=0)
std = trainX.std(axis=0)
trainX = (trainX - mean) / std
validX = (validX - mean) / std
```

Problem 2. Discuss which attribute has the greatest impact on income prediction accuracy, visualize the result and explain it.

- 下圖為將 Logistic Regression 的 weights 絕對值對應 columns 的視覺化長條圖。因為所有 features 總共有 100 多個 (after one hot encoding)，全部畫出來的話圖會太擠所以這邊只秀出前 15 大的 absolute weight
- 根據結果圖，可以看到前三名依序是 capital-gain, martial-status_Married-civ-spouse and martial-status_Never-married。而第一名的 capital-gain 遠遠超過其他的，可見

階級複製的威力



Problem 3. Discuss the impact of regularization on income prediction accuracy

- 下圖為 regularization 與 prediction accuracy 的比較圖，其中橫軸 alpha 是 regularization 的超參數，表示正則化的比例，而縱軸則是模型準確率。
- 可以看到隨著 alpha 增加但是 training & validation accuracy 都是持續在下降的，代表這個 case 可能不是很適合加上 regularization

